

Phase 1.5 Status Review

✓ Already Implemented (per scripts & progress log)

Level Save/Load

- **Save JSON:** `builder_mode.gd::_on_save_pressed()` writes timestamped JSON into `user://coils/`.
- **Load JSON:** `builder_mode.gd::_on_load_file_selected()` parses JSON, calls `CoilIO.apply_coil`, restores biomass.
- **Autosave:** Playtest triggers `_autosave_playtest.json` snapshot.
- **ExploreMode:** Implements `_load_from_coil()` and `_rebuild_layer_from_json()` for exact reconstruction of levels.

Builder-Explore Roundtrip

- **Hand-off:** `CoilSession.start_playtest(data)` stores snapshot and changes scene.
- **Return:** `crawler.gd::_win_and_return()` calls `CoilSession.return_to_builder()`.
- **Restore:** `builder_mode.gd::_restore_pending_coil_if_any()` re-applies coil snapshot.

Validation Integration

- **Validate** button → always strict, shows popup with messages, spawn, heart coords.
- **Playtest** validation → allows bypass of biomass only if DevMode + IgnoreBiomass enabled.

UI Hooks

- Buttons for **Validate**, **Save**, **Load**, **Playtest** are all wired.
- Status label updated on all key actions.
- Biomass label recalculates and displays warnings if over cap.

Schema & Metadata

- JSON schema includes `meta` (biomass used/cap, tileset) + `layers` arrays.
- Roundtrip format stable and already used by Explore.

☀ Still To Do in 1.5

1. Save Policy

2. Decide strict save (validate before saving) vs. draft save (allow invalid but flag `Validated: false`).

3. Player Profile Persistence

4. `player_profile.json` needed for:
 - Upgrade unlocks/toggles (from `upgrade_controller.gd`).
 - Resource counters (Nutrient, Sporeprint).

5. No profile system yet.

6. Publish Stub (Local)

7. Add **Publish** button that copies JSON to `/Published/`.

8. Stub only, no metadata or backend.

9. Level Browser QoL (Optional)

10. File dialog works, but design docs suggest lightweight browser UI for saved/published levels.

11. Error Handling / UX Polish

12. Status messages exist, but malformed JSON or open errors could use modal popup.

13. Schema Documentation

14. Write a short README with keys and guarantees to keep consistency across Builder/Explore/Tools.



Summary

- Core **save/load/roundtrip** is implemented and functional.
- Phase 1.5 is **90% done**.
- Remaining work: finalize save policy, add **profile persistence**, and stub **Publish** button to fully close the loop.